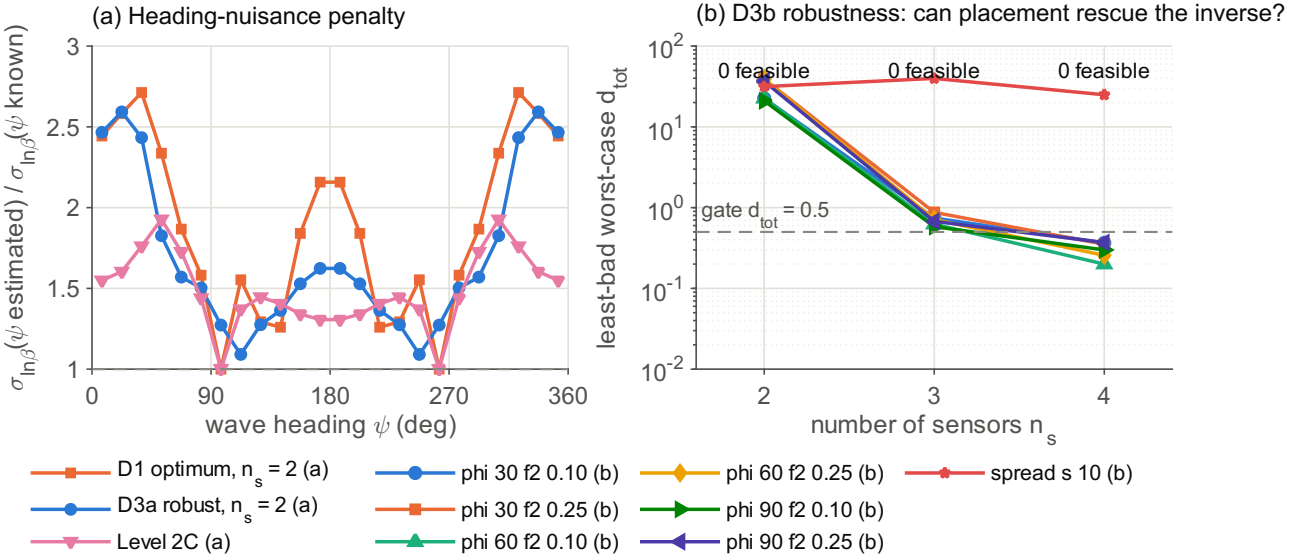

(a) Heading-nuisance penalty
$\sigma_{\ln\beta}(\psi$ estimated) / $\sigma_{\ln\beta}(\psi$ known)
3
2.5
2
1.5
1
0
90
180
270
360
wave heading ψ (deg)
(b) D3b robustness: can placement rescue the inverse?
least-bad worst-case d_{tot}
10^{2}
10^{1}
10^{0}
10^{-1}
10^{-2}
0 feasible
0 feasible
0 feasible
gate d_{tot} = 0.5
2
3
4
number of sensors n_s
D1 optimum, n_s = 2 (a)
D3a robust, n_s = 2 (a)
Level 2C (a)
phi 30 f2 0.10 (b)
phi 30 f2 0.25 (b)
phi 60 f2 0.10 (b)
phi 60 f2 0.25 (b)
phi 90 f2 0.10 (b)
phi 90 f2 0.25 (b)
spread s 10 (b)